

A One-Column Counterexample to the Reverse Toeplitz-Algebra Implication

Candidate solution packet for arXiv:2510.19589
agent_lane.04 (GPT5.6)

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Status. Candidate counterexample, likely valid, awaiting human review.

1 Source question

The source is Békollè, Défo, and Tchoundja, *Analysis of Toeplitz Operators with BMO_α^1 operator-valued symbols on ℓ^2 -valued Bergman spaces*, arXiv:2510.19589 (2025) [1]. Section 13, page 41 asks whether

$$b, b^* \in BMO_\alpha^1(\mathbb{B}_n, \mathcal{L}(\ell^2, \ell^1 \cap \ell^2)), \quad \tilde{b} \in L^\infty(\mathbb{B}_n, \mathcal{L}(\ell^2)),$$

together with

$$\lim_{d \rightarrow \infty} \|T_b Q_d\| = 0,$$

implies $T_b \in \mathcal{T}_{L_{\text{fin}}^\infty}$. Here $Q_d = M_{I(d)}$ is the orthogonal projection of ℓ^2 onto $\overline{\text{span}}\{e_j : j > d\}$, acting pointwise on $A_\alpha^2(\mathbb{B}_n, \ell^2)$.

13. AN OPEN QUESTION

For $d = 1$, Xia [15] proved the reverse implication: if S is sufficiently localized (and even weakly localized), then $S \in \mathcal{T}_\alpha^1$. It would be interesting to extend this result to the infinite dimension. More specifically, for Toeplitz operators, if $S = T_b$ such that $b, (b(\cdot))^*_{\mathcal{L}(\ell^2)} \in BMO_\alpha^1(\mathbb{B}_n, \mathcal{L}(\ell^2, \ell^1 \cap \ell^2))$ and $\tilde{b} \in L^\infty(\mathbb{B}_n, \mathcal{L}(\ell^2))$, does one have $T_b \in \mathcal{T}_{L_{\text{fin}}^\infty}$ whenever $\lim_{d \rightarrow \infty} \|T_b M_{I(d)}\|_{\mathcal{L}(A_\alpha^2(\mathbb{B}_n, \ell^2))} = 0$? This would also be an infinite dimensional generalization of the result of Sadeghi-Zorboska (Theorem 1.10 above).

Figure 1: The open question in Section 13, page 41 of the source paper.

2 Proof intuition

The hypothesis controls only input-coordinate tails. We therefore choose a symbol having exactly one nonzero input column; then $T_b Q_d = 0$ automatically for every $d \geq 1$. That column is distributed among infinitely many output coordinates using pairwise disjoint angular sectors of the ball. Normalized Bergman kernels concentrate at boundary points, so each sector produces a scalar Toeplitz component of norm one. Hence no output-coordinate tail can decay. This is incompatible with $\mathcal{T}_{L_{\text{fin}}^\infty}$, because its finite-symbol words have finite row and column support, and norm closure preserves two-sided tail decay.

3 Definitions

Let P_α be the scalar weighted Bergman projection, acting coordinatewise on $L_\alpha^2(\mathbb{B}_n, \ell^2)$. For a bounded operator-valued symbol b , set $T_b = P_\alpha M_b$. The normalized scalar kernel is

$$k_a^\alpha(w) = \frac{(1 - |a|^2)^{(n+1+\alpha)/2}}{(1 - \langle w, a \rangle)^{n+1+\alpha}}.$$

Write $P_d = I - Q_d$ for projection onto the first d coordinates. The algebra $\mathcal{T}_{L_{\text{fin}}^\infty}$ is the operator-norm closure of finite sums of finite products of Toeplitz operators whose symbols have only finitely many nonzero matrix entries, all bounded [1].

We use either standard norm on $\ell^1 \cap \ell^2$ (the sum or maximum of the two norms); the construction is bounded for both.

4 The counterexample

Lemma 1 (A necessary two-sided tail condition). *If $S \in \mathcal{T}_{L_{\text{fin}}^\infty}$, then*

$$\|SQ_d\| \longrightarrow 0 \quad \text{and} \quad \|Q_dS\| \longrightarrow 0.$$

Proof. If $u \in L_{\text{fin}}^\infty$, choose N larger than every row and column index on which u has a nonzero matrix entry. Since P_α acts coordinatewise,

$$Q_N T_u = 0, \quad T_u Q_N = 0.$$

For a finite product of such Toeplitz operators, the leftmost factor gives finite row support and the rightmost factor gives finite column support. Thus every finite sum of finite products has both tails identically zero once d is large enough. If S is an operator-norm limit of such operators, an ε -approximation immediately gives $\limsup_d \|Q_dS\| \leq \varepsilon$ and $\limsup_d \|SQ_d\| \leq \varepsilon$. Let $\varepsilon \downarrow 0$. $\square$

Lemma 2 (Disjoint sectors with norm-one Toeplitz operators). *There are pairwise disjoint Borel sets $E_1, E_2, \dots \subset \mathbb{B}_n$ such that*

$$\|T_{\chi_{E_j}}\| = 1 \quad (j \geq 1).$$

Proof. Choose pairwise disjoint relatively open caps C_j of the unit sphere and a point $\zeta_j \in C_j$. Let E_j be the radial sector consisting of nonzero $w \in \mathbb{B}_n$ whose direction $w/|w|$ lies in C_j . These sets are disjoint, and E_j contains a relative neighborhood of ζ_j in the closed ball.

The probability measures

$$d\mu_{r\zeta_j}(w) = |k_{r\zeta_j}^\alpha(w)|^2 d\nu_\alpha(w)$$

converge weakly to the point mass at ζ_j as $r \uparrow 1$. For completeness, if V is any relative neighborhood of ζ_j , compactness gives $c_V > 0$ such that

$$|1 - r\langle w, \zeta_j \rangle| \geq c_V \quad (w \in \overline{\mathbb{B}_n} \setminus V)$$

for all r sufficiently close to one. Therefore

$$\mu_{r\zeta_j}(\mathbb{B}_n \setminus V) \leq C_V(1 - r^2)^{n+1+\alpha} \longrightarrow 0.$$

Taking $V \cap \mathbb{B}_n \subset E_j$ yields $\widetilde{\chi_{E_j}}(r\zeta_j) \rightarrow 1$. Since $T_{\chi_{E_j}}$ is a positive contraction and

$$\langle T_{\chi_{E_j}} k_a^\alpha, k_a^\alpha \rangle = \widetilde{\chi_{E_j}}(a),$$

its norm is exactly one. $\square$

Theorem 1. *For every $n \geq 1$ and $\alpha > -1$, the question in Section 13 of arXiv:2510.19589 has a negative answer. More strongly, there is a bounded strongly measurable symbol*

$$b : \mathbb{B}_n \rightarrow \mathcal{L}(\ell^2, \ell^1 \cap \ell^2)$$

whose Hilbert-space adjoint has the same property, such that

$$b, b^* \in BMO_\alpha^1(\mathbb{B}_n, \mathcal{L}(\ell^2, \ell^1 \cap \ell^2)), \quad \|\tilde{b}\|_{L^\infty(\mathcal{L}(\ell^2))} \leq 1,$$

and

$$T_b Q_d = 0 \quad (d \geq 1), \quad \|Q_d T_b\| = 1 \quad (d \geq 1).$$

Consequently $T_b \notin \mathcal{T}_{L_{\text{fin}}^\infty}$.

Proof. For the sectors above, define

$$v(w) = \sum_{j \geq 1} \chi_{E_j}(w) e_j, \quad b(w)x = \langle x, e_1 \rangle v(w).$$

Because the E_j are disjoint, $v(w)$ is either zero or one standard basis vector. Hence b is strongly measurable, takes only countably many values, and is uniformly bounded from ℓ^2 into $\ell^1 \cap \ell^2$. Its adjoint is

$$b(w)^* x = \langle x, v(w) \rangle e_1,$$

and has the same boundedness and measurability properties.

Every bounded Banach-valued symbol c belongs to the paper's BMO_α^1 class: from its definition,

$$\sup_{z \in \mathbb{B}_n} \int_{\mathbb{B}_n} \|c \circ \varphi_z(w) - \tilde{c}(z)\| d\nu_\alpha(w) \leq 2\|c\|_\infty.$$

Thus both required BMO_α^1 hypotheses hold.

Set

$$p_j(z) = \widetilde{\chi_{E_j}}(z) = \int_{E_j} |k_z^\alpha(w)|^2 d\nu_\alpha(w).$$

Then $p_j(z) \geq 0$ and disjointness gives $\sum_j p_j(z) \leq 1$. Moreover,

$$\tilde{b}(z)x = \langle x, e_1 \rangle \sum_{j \geq 1} p_j(z) e_j.$$

The output vector has ℓ^2 norm at most its ℓ^1 norm, which is at most one. Hence $\|\tilde{b}(z)\|_{\mathcal{L}(\ell^2)} \leq 1$ uniformly in z .

For $d \geq 1$, $b(w)Q_d = 0$ pointwise, so $T_b Q_d = 0$. On the other hand, for a scalar $f \in A_\alpha^2(\mathbb{B}_n)$,

$$T_b(f e_1) = (T_{\chi_{E_1}} f, T_{\chi_{E_2}} f, \dots).$$

Fix $j > d$. The j th component implies

$$\|Q_d T_b\| \geq \|T_{\chi_{E_j}}\| = 1.$$

Since b is pointwise contractive, $\|T_b\| \leq 1$, so equality holds. Lemma 1 now shows that $T_b \notin \mathcal{T}_{L_{\text{fin}}^\infty}$. $\square$

By Theorem 1.19 of the source paper, this T_b is also sufficiently localized. Thus the example disproves the proposed infinite-dimensional reverse implication even inside the source's localized Toeplitz class.

5 Verification, scope, and novelty

The proof is exact and uses no computation. The construction satisfies the question with the stronger identity $T_b Q_d = 0$ for every $d \geq 1$. It does not challenge Proposition 1.16 of the source: that proposition additionally assumes the missing left-tail condition $\|Q_d T_b\| \rightarrow 0$, which fails here as strongly as possible.

A bounded novelty search on 26 August 2026 used the exact arXiv id and title, the notation $\mathcal{T}_{L^\infty_{\text{fin}}}$, exact phrases from Section 13, and combinations of operator-valued BMO, Toeplitz algebra, and reverse implication. It found the source paper, mirrors, and summaries, but no later explicit answer. This is not an exhaustive bibliographic guarantee.

Human-review recommendation. Treat Theorem 1 as a high-priority candidate counterexample. In particular, verify the bounded Banach-valued BMO observation and Lemma 1; neither step hides a compactness or uniformity assumption.

References

- [1] D. Békollè, H. O. Défo, and E. L. Tchoundja, *Analysis of Toeplitz Operators with BMO_α^1 operator-valued symbols on ℓ^2 -valued Bergman spaces*, arXiv:2510.19589, 2025.